KOUSHIK VASA

AI Engineer | Agentic AI & LLM Systems

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PROFESSIONAL SUMMARY

AI Engineer with **2+** years building production LLM systems end-to-end, from multi-agent orchestration and RAG pipelines to evaluation and latency optimization. Raised agent task success from 62% to 84%, cut hallucinations by 25%, and brought responses under 400ms as the sole AI engineer at Bridgenow.ai. Most recently the sole backend engineer on MedEQ, an AI medical education platform, building Claude-driven clinical simulations, an FDA drug-label RAG pipeline with full source lineage, and an ACCME-aligned accreditation system.

TECHNICAL SKILLS

Generative AI & LLM Systems: Agentic AI, Multi-Agent Orchestration, Tool Calling, RAG (Hybrid, Re-Ranking), Embeddings, Semantic Search, Prompt Engineering, LLM Evaluation

LLM Frameworks & Platforms: Anthropic Claude, OpenAI API, DSPy, LangGraph, LangChain, Hugging Face, Vapi, Twilio

ML & NLP: Scikit-learn, XGBoost, PyTorch, NLP, Feature Engineering, Model Evaluation, Data Pipelines (ETL)

Backend & Infrastructure: Azure (ML, AI Search, Databricks), AWS (S3, Lambda), FastAPI, Node.js, Supabase (PostgreSQL), pgvector, Cloudflare Workers, Docker, Vercel, GitHub Actions, OAuth

Frontend: Next.js, React, Vue.js, TypeScript, Tailwind CSS

Languages: Python, SQL, TypeScript, Bash

EDUCATION

Master of Science in Computer Science (Machine Learning) — GPA 3.87 *Jan 2024 – Dec 2025*

George Mason University, Fairfax, VA

PROFESSIONAL EXPERIENCE

AI Engineer *Apr 2026 – Present*

MedEQ | USA

- Sole backend engineer on MedEQ, an AI-powered continuing medical education platform where clinicians practice against Claude-driven patient simulations; owned architecture across FastAPI, Cloudflare Workers, and Supabase (Postgres + pgvector).
- Architected a 7-module CME accreditation system (identity & NPI verification, idempotent batch credit engine, typed/spoken reflection workflow, certificates, audit reporting) built to ACCME requirements with an external accreditation partner.
- Rebuilt the FDA drug-label RAG ingestion pipeline with end-to-end source lineage across all 15 FDA label sections, making every clinical answer traceable to its regulatory source.
- Built a data-driven clinical simulation engine with severity-based patient vitals logic and orchestration observability across multi-call LLM flows, making AI behavior auditable for clinical reviewers.

Generative AI Engineer *Oct 2025 – Apr 2026*

Bridgenow.ai | USA

- Sole AI engineer on Bridgette, an agentic assistant for a B2B expert marketplace, replacing a 10+ step workflow (search → filter → invite → negotiate) with a single conversational interface; owned agent architecture, backend APIs, and frontend streaming.
- Architected a production dual-agent system (companies vs. experts) using Claude and DSPy, with 8 Zod-validated tool schemas powering expert matching, engagement creation, and earnings analysis.
- Raised end-to-end task success from ~62% to ~84% and cut hallucinated or incomplete responses by ~25% through validation layers and fallbacks, evaluated on 500+ test queries.
- Reduced average response time from ~1.2s to under 400ms through prompt optimization and query-level caching.

AI Engineer *Jan 2023 – Dec 2023*

Cyware Labs Pvt Ltd | India

- Built Python pipelines to parse, normalize, and sanitize raw security logs into structured, high-signal datasets feeding downstream ML anomaly detection models.
- Integrated threat-intelligence dashboards with backend AI/ML services, resolving 150+ integration issues to deliver model-driven risk scores to SOC analysts in real time.
- Led the Vue 2 → Vue 3 migration of threat-intelligence dashboards, improving rendering performance of live predictive risk-scoring views by 30%.
- Improved product stability by 40% through systematic bug resolution in an Agile workflow (Git, Bitbucket, Jira).

PROJECTS

Sonus: Clinical Voice Intake Agent | Next.js, TypeScript, Vapi, Claude Sonnet, Tavily

- Hackathon winner: built a voice AI agent that runs structured pre-consultation interviews with patients, capturing symptoms and clinical context before the visit.
- Generated physician-ready SOAP notes from the conversation with Claude Sonnet, cutting manual intake documentation.
- Deployed a live demo on Vercel with an end-to-end voice → transcript → clinical note flow.

ClearCare: AI Medicare Cost Navigator | Next.js, FastAPI, LangGraph, GPT-4o, ElevenLabs, Supabase

- Hackathon winner: built an 8-node LangGraph multi-agent pipeline that maps patient symptoms to likely procedures and calculates real out-of-pocket Medicare costs.
- Grounded cost estimates in 2.8M+ CMS NPI provider records, returning provider-specific pricing rather than generic averages.
- Added voice interaction via ElevenLabs so patients can ask cost questions conversationally.

CitationSleuth: RAG-Based Fact Verification System | Python, Neo4j, Embeddings, RAG

- Built dual-layer LLM validation system combining semantic retrieval and Neo4j graph traversal to verify generated claims.
- Improved verification precision by linking embedding-based evidence retrieval with graph-based relationship validation.
- Developed real-time interface surfacing low-confidence or unsupported outputs before downstream usage.

CERTIFICATIONS

- Generative AI LLMs — NVIDIA (Jan 2026)
- Microsoft Fabric Analytics — Microsoft (Jan 2026)
- Agentforce — Salesforce (Dec 2025)
- OCI AI Foundations Associate — Oracle (Oct 2024)
- Generative AI Fundamentals — Databricks (Aug 2024)
- Prompt Engineering for Everyone — IBM (Aug 2024)